

Supplementary Material: Sensing Which Modality Matters: Evidence-Gated Regularization for Robust VLA Policies

Anonymous Author(s)

1 This appendix supplements the main paper with implementation details, benchmark construction de-
2 tails, statistical analyses, and a theoretical derivation. Section A reports hyperparameters. Section B
3 reports filter statistics, threshold values, and the sim state replay procedure. Section C formally de-
4 fines the Suite 1 metric Δ and reports standard errors, confidence intervals, and significance tests
5 for all numbers in the main paper. Section D proves that the importance-sampled estimators used
6 during training are unbiased.

7 A Implementation Details

8 The values in this section specify the configuration used to train EGR on BEHAVIOR-1K in sim-
9 ulation and the synthetic corruption operator used to construct the Suite 1 and Suite 2 evaluations.
10 The configuration is identical across all EGR simulation runs reported in the main text.

11 Table 1 lists the hyperparameters introduced by EGR itself: the two loss weights, the hinge thresh-
12 olds, the evidence-preprocessing constants, and a flag governing how the per-camera mask channel
13 is treated under corruption.

Symbol	Value	Description
λ_{inv}	0.2	weight on $\mathcal{L}_{\text{inv}}$ in the total loss
λ_{suff}	0.1	weight on $\mathcal{L}_{\text{suff}}$ (denoted <code>lambda_solo</code> in the released code)
τ_{low}	0.2	hinge knee below which w_{inv} activates
τ_{high}	0.5	hinge knee above which w_{suff} activates
α	0.3	weight of the gated interaction-object contribution in the per-camera score $s_{t,m}$
ϵ_{focal}	10^{-3}	interaction-gate threshold on focal-object area $a_{t,m}^{\text{focal}}$
ϵ_{global}	10^{-3}	global-gate threshold on $\max_m s_{t,m}$
ϵ	10^{-8}	numerical stabilizer in cross-camera normalization
<code>set_mask</code>	False	corrupted cameras retain <code>image_mask=True</code> so patch tokens stay valid

Table 1: EGR-specific hyperparameters.

14 Two distinct corruption operators are used. The training-time operator $\mathcal{C}_m$ injects random rectangu-
15 lar erasures during the computation of $\mathcal{L}_{\text{inv}}$ and $\mathcal{L}_{\text{suff}}$ (Table 2). The deployment-time operator used
16 in simulation evaluation applies a center-biased black erasure that approximates a large occlusion
17 (Table 3). Both operators sample rectangles independently per camera.

Parameter	Setting
Number of rectangles	uniform integer in $\{2, 3, 4, 5\}$
Total erased-area fraction	Uniform(0.30, 0.80)
Per-rectangle aspect ratio	$\exp(\text{Uniform}(-1, 1)) \approx [0.37, 2.72]$
Rectangle placement	top-left position uniform over image
Fill value	per-pixel Uniform($-1, 1$) noise (images normalized to $[-1, 1]$)
Per-camera independence	yes

Table 2: Training-time corruption operator $\mathcal{C}_m$ used inside $\mathcal{L}_{\text{inv}}$ and $\mathcal{L}_{\text{suff}}$.

Parameter	Setting
Number of rectangles	uniform integer in $\{1, 2, 3\}$
Total erased-area fraction	Uniform(0.30, 0.70)
Per-rectangle aspect ratio	Uniform(0.5, 2.0)
Rectangle placement	Gaussian about image center with $\sigma = 0.10 W$ and $0.10 H$, clipped to fit
Fill value	0 (black)
Per-camera independence	yes

Table 3: Sim-deployment corruption operator used in Suite 1 and Suite 2 to instantiate the NoUseless, UsefulOnly, and SingleUseful regimes.

Table 4 lists the training-loop configuration. All EGR simulation runs warm-start from the $\pi_{0.5}$ checkpoint provided by [?], pre-trained on the 12 BEHAVIOR-1K Challenge tasks; only LoRA adapter weights are updated.

Setting	Value
Batch size	288
Optimizer	AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-8}$
Peak learning rate	2.5×10^{-5}
LR schedule	cosine decay, 1,000-step warmup, decay over 50,000 steps, end LR 0
Total training steps	50,000
Weight decay	10^{-10}
Gradient clipping	global-norm clip at 1.0
EMA	disabled
Numerical precision	bfloat16 model; float32 loss and parameter accumulators
Hardware	$2 \times$ NVIDIA B200, FSDP across both devices
Dataloader workers	8
PaliGemma backbone	gemma_2b_lora (LoRA on the 2B base)
Action expert	gemma_300m_lora (LoRA on the 300M base)
Action horizon	32
Trainable parameters	LoRA adapters only; base weights frozen

Table 4: Training-loop configuration for EGR simulation runs.

B Benchmark Construction

B.1 Filter Pipeline Statistics

The benchmark curation pipeline starts from the 11 BEHAVIOR-1K Challenge tasks that yield usable annotations. Task 34 is excluded upstream because its annotation is truncated, removing 4 candidate segments before the evidence pipeline is run. The 11 retained tasks contribute 118 candidate segments, 39 navigation (NAV) and 79 manipulation (MAN). Three vision-evidence filters reduce this pool to the 47 segments evaluated in the main text. Filter 1 retains segments whose per-frame global gate $G_t = 1$ on at least 80% of frames, removing segments in which no camera observes the focal object on a typical frame. Filter 2 retains segments whose head camera is labeled useful, restricting evaluation to scenarios where the head view is informative. Filter 3 enforces a clean wrist-camera contrast: MAN segments are kept only when exactly one wrist is useless (so that the contrast between the useful wrist and the useless wrist is unambiguous), and NAV segments are kept when at least one wrist is useless (a looser criterion, since wrists are typically uninformative during locomotion). Table 5 reports the segment count after each filter, broken out by skill type.

Stage	All	NAV	MAN
Input (11 evaluation tasks)	118	39	79
After Filter 1 (segment-level global-gate pass)	103	25	78
After Filter 2 (head-camera useful)	99	25	74
After Filter 3 (wrist-camera ablation criterion)	47	11	36

Table 5: Segment counts through the three vision-evidence filters.

B.2 Threshold Values

Two thresholds operate at the per-frame level and drive the per-camera useful/useless classification: ϵ_{global} defines when a frame carries evidence at all, and `useless_threshold` defines when a frame counts as not seeing the focal object on a given camera. The remaining two thresholds aggregate frame-level outcomes to segment-level decisions: `g_threshold` on the fraction of frames passing the global gate, and `frac_threshold` on the fraction of frames below `useless_threshold`. A camera m is labeled useless on a segment if at least 75% of its frames satisfy $E_{t,m} < 0.01$, and useful otherwise. Table 6 lists each threshold, its numeric value, and which filter consumes it.

Threshold	Value	What it gates	Consumed by
ϵ_{global}	10^{-3}	per-frame global gate, $G_t = 1$ iff $\max_m s_{t,m} > 10^{-3}$	Filter 1 (defines G_t)
<code>g_threshold</code>	0.80	minimum fraction of frames with $G_t = 1$ for a segment to pass	Filter 1
<code>useless_threshold</code>	0.01	per-frame, per-camera cutoff on $E_{t,m}$ below which the frame counts as not seeing the focal object	Filter 2 and Filter 3
<code>frac_threshold</code>	0.75	fraction of frames below <code>useless_threshold</code> required to label a camera useless on the segment	Filter 2 and Filter 3

Table 6: Threshold values used in the filter pipeline.

B.3 Sim State Replay Procedure

The official BEHAVIOR-1K HDF5 stores per-frame state as a slot-keyed serialized tensor; loading it into a Suite 2 evaluation scene with a different object set mis-aligns the slots, and a single-frame jump-in additionally loses hidden physics state such as joint velocities, contact constraints, and particle-system bookkeeping. We therefore replay each task’s first episode from frame 0 in VISUAL_ONLY mode and, at each segment start frame, call `og.sim.dump_state(serialized=False)` to save a name-keyed snapshot covering all object poses, robot and gripper joint configurations, and particle-system state. At evaluation time, Suite 2 loads the snapshot directly into the evaluation scene and runs one zero-gravity settling step so PhysX re-establishes contact constraints before the policy begins acting.

C Metrics and Statistical Analysis

C.1 Definition of Δ for Suite 1

For each segment with relevant action dimensions $\mathcal{D}$ and frames $\mathcal{F}$,

$$\Delta = \underbrace{\frac{1}{|\mathcal{F}|} \sum_{t \in \mathcal{F}} \|\pi_{\theta}(o_t^{\text{corrupt}}) - a_t^*\|_{\mathcal{D}}^2}_{\text{MSE}_{\text{corrupt}}} - \underbrace{\frac{1}{|\mathcal{F}|} \sum_{t \in \mathcal{F}} \|\pi_{\theta}(o_t^{\text{full}}) - a_t^*\|_{\mathcal{D}}^2}_{\text{MSE}_{\text{full}}}, \quad (1)$$

where a_t^* is the ground-truth action chunk and $\mathcal{D}$ selects the robot base dimensions for NAV segments and the torso plus both arm dimensions for MAN segments.

58 C.2 Standard Errors and Confidence Intervals

59 This subsection reports per-cell uncertainty for every number used in Tables 1 and 2 and in Fig. 4
60 of the main paper. Aggregated means use $SE = SD/\sqrt{n}$ across the natural sampling unit: segments
61 for the simulation suites and tasks for the real-robot platforms. Binomial rates additionally report the
62 Wilson 95% interval over the pooled trial count. Suite 1 Ratios additionally report a 95% bootstrap
63 interval from 10,000 paired resamples over segments. All computations use a fixed seed.

64 **Suite 1.** Each cell reports the Full and corrupt MSE means with standard errors and the Ratio with
65 a bootstrap 95% interval; the EGR row has the lowest Ratio in every panel.

Panel	Method	n	Full MSE (mean $\pm$ SE)	Corrupt MSE (mean $\pm$ SE)	Ratio % (mean $\pm$ SE) [95% CI]
NAV / SingleUseful	vanilla	11	0.0645 $\pm$ 0.0110	0.0819 $\pm$ 0.0125	+26.90 $\pm$ 9.84 [+14.1, +45.0]
NAV / SingleUseful	ModDrop	11	0.0872 $\pm$ 0.0141	0.1160 $\pm$ 0.0188	+33.12 $\pm$ 23.23 [+12.5, +67.2]
NAV / SingleUseful	EGR	11	0.0796 $\pm$ 0.0125	0.0894 $\pm$ 0.0140	+12.21 $\pm$ 3.53 [+6.3, +18.2]
MAN / NoUseless	vanilla	36	0.0020 $\pm$ 0.0002	0.0023 $\pm$ 0.0002	+13.97 $\pm$ 4.13 [+5.2, +24.5]
MAN / NoUseless	ModDrop	36	0.0024 $\pm$ 0.0002	0.0026 $\pm$ 0.0002	+10.55 $\pm$ 3.37 [+5.5, +15.9]
MAN / NoUseless	EGR	36	0.0023 $\pm$ 0.0003	0.0024 $\pm$ 0.0003	+4.80 $\pm$ 2.63 [-0.0, +10.8]
MAN / UsefulOnly	vanilla	36	0.0020 $\pm$ 0.0002	0.0052 $\pm$ 0.0008	+163.64 $\pm$ 21.57 [+110.2, +223.6]
MAN / UsefulOnly	ModDrop	36	0.0024 $\pm$ 0.0002	0.0057 $\pm$ 0.0008	+139.94 $\pm$ 16.59 [+100.8, +180.9]
MAN / UsefulOnly	EGR	36	0.0023 $\pm$ 0.0003	0.0051 $\pm$ 0.0007	+124.14 $\pm$ 18.07 [+76.2, +177.3]

Table 7: Suite 1 per-cell statistics. Standard errors are computed across segments; Ratio confidence intervals are from 10,000 paired bootstrap resamples over segments.

66 **Suite 2 (NAV).** Per-segment SR is a binomial proportion over $n = 20$ trials, and we report the
67 cross-segment mean $\pm$ SE together with a bootstrap 95% CI over $n = 11$ segments.

Condition	Method	n_{seg}	SR (mean $\pm$ SE)	95% CI
Full	vanilla	11	42.27 $\pm$ 14.78	[15.9, 70.0]
Full	ModDrop	11	40.00 $\pm$ 14.13	[15.5, 66.8]
Full	EGR	11	40.91 $\pm$ 14.47	[15.0, 68.2]
SingleUseful	vanilla	11	15.45 $\pm$ 9.28	[1.8, 34.5]
SingleUseful	ModDrop	11	15.45 $\pm$ 9.18	[1.8, 34.5]
SingleUseful	EGR	11	37.27 $\pm$ 13.76	[13.6, 63.6]

Table 8: Suite 2 NAV success rate per cell. Means and SEs are computed across $n_{\text{seg}} = 11$ segments; CIs are from 10,000 bootstrap resamples over segments.

68 **Suite 2 (MAN).** For each cell, the mean is computed across $n_{\text{seg}} = 36$ MAN segments; each per-
69 segment value summarizes 20 trials. We report SE and bootstrap 95% CI across segments for all
70 three metrics.

Condition	Method	SR (%)		EC (%)		EE distance (m)	
		mean $\pm$ SE	95% CI	mean $\pm$ SE	95% CI	mean $\pm$ SE	95% CI
Full	vanilla	12.50 $\pm$ 3.77	[6.0, 20.4]	52.78 $\pm$ 6.15	[40.8, 64.6]	0.249 $\pm$ 0.049	[0.16, 0.35]
Full	ModDrop	7.50 $\pm$ 3.17	[2.5, 14.6]	40.97 $\pm$ 6.24	[28.9, 52.9]	0.287 $\pm$ 0.048	[0.20, 0.39]
Full	EGR	16.39 $\pm$ 4.10	[8.9, 24.7]	50.00 $\pm$ 5.95	[38.5, 61.4]	0.257 $\pm$ 0.049	[0.17, 0.36]
NoUseless	vanilla	9.44 $\pm$ 3.41	[3.8, 16.8]	47.22 $\pm$ 6.01	[35.7, 58.6]	0.264 $\pm$ 0.045	[0.18, 0.36]
NoUseless	ModDrop	6.81 $\pm$ 3.05	[1.9, 13.3]	38.89 $\pm$ 5.96	[27.5, 50.4]	0.292 $\pm$ 0.047	[0.21, 0.39]
NoUseless	EGR	16.53 $\pm$ 4.21	[9.2, 25.1]	53.89 $\pm$ 6.07	[41.9, 65.6]	0.246 $\pm$ 0.046	[0.16, 0.34]
UsefulOnly	vanilla	2.78 $\pm$ 2.78	[0.0, 8.3]	22.78 $\pm$ 6.29	[11.2, 35.1]	0.447 $\pm$ 0.059	[0.34, 0.57]
UsefulOnly	ModDrop	2.78 $\pm$ 2.78	[0.0, 8.3]	22.36 $\pm$ 6.34	[10.8, 35.0]	0.470 $\pm$ 0.064	[0.35, 0.60]
UsefulOnly	EGR	6.11 $\pm$ 3.05	[1.4, 12.9]	30.14 $\pm$ 6.01	[18.9, 42.2]	0.366 $\pm$ 0.052	[0.27, 0.47]

Table 9: Suite 2 MAN per-cell statistics for the three reported metrics. $n_{\text{seg}} = 36$ for every cell.

71 **Real bi-manual platform.** Each (task, condition, method) cell contains $n = 10$ trials, pooled
72 across 4 tasks to $N = 40$ trials per (condition, method). We report the pooled rate with its Wilson
73 95% interval over the 40 trials, together with the cross-task SE over $n_{\text{task}} = 4$.

Condition	Baseline			EGR		
	Pooled n/N	Wilson 95% CI	Cross-task SE	Pooled n/N	Wilson 95% CI	Cross-task SE
Clean	28/40	[0.55, 0.82]	0.70 ± 0.07	32/40	[0.65, 0.90]	0.80 ± 0.07
NoUseless	21/40	[0.37, 0.67]	0.53 ± 0.05	36/40	[0.77, 0.96]	0.90 ± 0.07
UsefulOnly	10/40	[0.14, 0.40]	0.25 ± 0.10	29/40	[0.57, 0.84]	0.73 ± 0.10
RealDistractor	12/40	[0.18, 0.45]	0.30 ± 0.15	34/40	[0.71, 0.93]	0.85 ± 0.09

Table 10: Real bi-manual platform SR per condition. The pooled count n aggregates 4 tasks $\times$ 10 trials; the cross-task SE is over $n_{\text{task}} = 4$ task-level SRs.

74 **Real tactile platform.** SR per (condition, method) cell pools the 2 tasks $\times$ 10 trials into $N = 20$
75 trials.

Condition	Baseline		EGR	
	Pooled n/N	Wilson 95% CI	Pooled n/N	Wilson 95% CI
Clean	18/20	[0.70, 0.97]	17/20	[0.64, 0.95]
SingleUseful	8/20	[0.22, 0.61]	18/20	[0.70, 0.97]
RealDistractor	11/20	[0.34, 0.74]	14/20	[0.48, 0.85]

Table 11: Real tactile platform SR per condition, pooled across the 2 tasks.

76 C.3 Significance Tests

77 For the simulation suites we use the one-sided paired Wilcoxon signed-rank test on per-segment
78 values, with EGR as the test arm and one of the two baselines as the reference. The alternative
79 hypothesis follows the metric’s direction: for Suite 1 Ratio we test $\text{EGR } \Delta < \text{baseline } \Delta$ where
80 $\Delta = \text{MSE}_{\text{corrupt}} - \text{MSE}_{\text{full}}$; for Suite 2 SR and EC we test $\text{EGR} > \text{baseline}$; for Suite 2 EE
81 distance we test $\text{EGR} < \text{baseline}$. For the real-robot platforms we use Fisher’s exact test (two-sided)
82 on the pooled trial counts. Significance markers throughout: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
83 EGR is designed to mitigate corruption rather than to improve performance under Clean inputs, so
84 non-significance on Full and Clean conditions is consistent with the design intent.

85 **Suite 1.** EGR’s per-segment Δ is significantly smaller than vanilla in all three panels and smaller
86 than ModDrop in two of the three; the MAN NoUseless comparison against ModDrop is marginal
87 ($p = 0.078$).

Panel	Comparison	n	W	p
NAV / SingleUseful	EGR vs vanilla	11	10.0	0.0210*
NAV / SingleUseful	EGR vs ModDrop	11	10.0	0.0210*
MAN / NoUseless	EGR vs vanilla	36	201.0	0.0189*
MAN / NoUseless	EGR vs ModDrop	36	242.0	0.0784
MAN / UsefulOnly	EGR vs vanilla	36	185.0	0.0096**
MAN / UsefulOnly	EGR vs ModDrop	36	182.0	0.0084**

Table 12: Suite 1 paired Wilcoxon signed-rank tests on per-segment Δ . One-sided, EGR $\Delta <$ baseline Δ .

88 **Suite 2 (NAV).** With only 11 NAV segments the paired Wilcoxon attains its floor of $p = 2^{-4} =$
89 0.0625 under SingleUseful; all four segments on which the methods differ favor EGR, so the limi-
90 tation is statistical power rather than effect direction or consistency.

Condition	Comparison	n_{seg}	nonzero diffs	W	p
Full	EGR vs vanilla	11	3	1.0	0.8750
Full	EGR vs ModDrop	11	3	4.0	0.3750
SingleUseful	EGR vs vanilla	11	4	10.0	0.0625
SingleUseful	EGR vs ModDrop	11	4	10.0	0.0625

Table 13: Suite 2 NAV paired Wilcoxon. With only 11 segments and few nonzero differences, the smallest attainable p is $2^{-(\text{nonzero})}$. Under SingleUseful all 4 nonzero per-segment differences favor EGR, attaining the floor $p = 2^{-4} = 0.0625$ for both baseline comparisons.

91 **Suite 2 (MAN).** The Full-condition SR gain is significant against ModDrop ($p < 0.001$) and
92 marginal against vanilla ($p = 0.065$); it is a secondary effect, since EGR primarily targets the
93 corruption conditions, where NoUseless and UsefulOnly are significant against both baselines.

Metric	Condition	Comparison	p
SR	Full	EGR vs vanilla	0.0649
SR	Full	EGR vs ModDrop	0.0003***
SR	NoUseless	EGR vs vanilla	0.0055**
SR	NoUseless	EGR vs ModDrop	0.0002***
SR	UsefulOnly	EGR vs vanilla	0.0033**
SR	UsefulOnly	EGR vs ModDrop	0.0033**
EC	Full	EGR vs vanilla	0.8079
EC	Full	EGR vs ModDrop	0.0029**
EC	NoUseless	EGR vs vanilla	0.0186*
EC	NoUseless	EGR vs ModDrop	0.0001***
EC	UsefulOnly	EGR vs vanilla	0.0057**
EC	UsefulOnly	EGR vs ModDrop	0.0073**
EE distance	Full	EGR vs vanilla	0.7540
EE distance	Full	EGR vs ModDrop	0.0051**
EE distance	NoUseless	EGR vs vanilla	0.1990
EE distance	NoUseless	EGR vs ModDrop	0.0025**
EE distance	UsefulOnly	EGR vs vanilla	0.0000***
EE distance	UsefulOnly	EGR vs ModDrop	0.0000***

Table 14: Suite 2 MAN paired Wilcoxon signed-rank tests across $n_{\text{seg}} = 36$ segments. SR/EC use $H_1 : \text{EGR} > \text{baseline}$; EE distance uses $H_1 : \text{EGR} < \text{baseline}$.

94 **Real bi-manual.** All three corruption conditions (NoUseless, UsefulOnly, RealDistractor) are sig-
 95 nificant at $p < 0.001$ against the baseline, while Clean shows no significant difference, consistent
 96 with EGR not targeting clean performance.

Condition	Baseline n/N	EGR n/N	Odds ratio	p
Clean	28/40	32/40	1.71	0.4391
NoUseless	21/40	36/40	8.14	0.0004***
UsefulOnly	10/40	29/40	7.91	0.0000***
RealDistractor	12/40	34/40	13.22	0.0000***

Table 15: Real bi-manual Fisher’s exact test (two-sided) on pooled trial counts across the 4 tasks.

97 **Real tactile.** On RealDistractor the tactile platform pools only $N = 20$ trials and the EGR advan-
 98 tage (14/20 vs 11/20) does not reach significance ($p = 0.51$); the same RealDistractor condition on
 99 the bi-manual platform, with $N = 40$ trials, is strongly significant ($p < 0.001$, Table 15), indicating
 100 that the tactile result is underpowered rather than absent.

Condition	Baseline n/N	EGR n/N	Odds ratio	p
Clean	18/20	17/20	0.63	1.0000
SingleUseful	8/20	18/20	13.50	0.0022**
RealDistractor	11/20	14/20	1.91	0.5145

Table 16: Real tactile platform Fisher’s exact test (two-sided) on pooled SR. RealDistractor shows a positive point estimate that does not reach significance at $N = 20$ pooled trials.

101 D Unbiased Importance-Sampled Estimator for $\mathcal{L}_{\text{inv}}$ and $\mathcal{L}_{\text{suff}}$

102 The per-sample contribution of $\mathcal{L}_{\text{inv}}$ in Eq. (2) of the main paper is

$$\ell_b^{\text{inv}} = G_t \sum_{m \in \mathcal{M}} w_{\text{inv}}(E_{b,m}) \cdot \|v_\theta(o_b) - v_\theta(\hat{o}_b^{(m)})\|_2^2. \quad (2)$$

103 Let $W_b^{\text{inv}} = \sum_{m \in \mathcal{M}} w_{\text{inv}}(E_{b,m})$. When $W_b^{\text{inv}} > 0$, define the categorical distribution

$$p_{b,m}^{\text{inv}} = \frac{w_{\text{inv}}(E_{b,m})}{W_b^{\text{inv}}}, \quad \sum_{m \in \mathcal{M}} p_{b,m}^{\text{inv}} = 1. \quad (3)$$

104 Rewriting the per-sample loss using this distribution,

$$\ell_b^{\text{inv}} = G_t \cdot W_b^{\text{inv}} \cdot \sum_{m \in \mathcal{M}} p_{b,m}^{\text{inv}} \cdot \|v_\theta(o_b) - v_\theta(\hat{o}_b^{(m)})\|_2^2 = G_t \cdot W_b^{\text{inv}} \cdot \mathbb{E}_{m \sim p_b^{\text{inv}}} \left[\|v_\theta(o_b) - v_\theta(\hat{o}_b^{(m)})\|_2^2 \right]. \quad (4)$$

105 Our estimator draws a single sample $m_b^* \sim p_b^{\text{inv}}$ and forms

$$\hat{\ell}_b^{\text{inv}} = G_t \cdot W_b^{\text{inv}} \cdot \|v_\theta(o_b) - v_\theta(\hat{o}_b^{(m_b^*)})\|_2^2. \quad (5)$$

106 Taking expectation over m_b^* ,

$$\mathbb{E}_{m_b^* \sim p_b^{\text{inv}}} [\hat{\ell}_b^{\text{inv}}] = G_t \cdot W_b^{\text{inv}} \cdot \mathbb{E}_{m \sim p_b^{\text{inv}}} \left[\|v_\theta(o_b) - v_\theta(\hat{o}_b^{(m)})\|_2^2 \right] = \ell_b^{\text{inv}}. \quad (6)$$

107 The W_b^{inv} multiplier is the importance-sampling correction; without it the estimator would be biased
 108 downward by a factor of W_b^{inv} . When $W_b^{\text{inv}} = 0$, no modality on sample b triggers the invariance
 109 hinge; we then fall back to a uniform $1/|\mathcal{M}|$ prior over modalities for sampling, and the loss contri-
 110 bution remains zero through the W_b^{inv} multiplier. Summing $\hat{\ell}_b^{\text{inv}}$ over the batch gives the estimator
 111 $\hat{\mathcal{L}}_{\text{inv}}$ of Eq. (5) of the main paper, so $\mathbb{E}[\hat{\mathcal{L}}_{\text{inv}}] = \mathcal{L}_{\text{inv}}$. The derivation for $\hat{\mathcal{L}}_{\text{suff}}$ is identical, with w_{suff}
 112 and W_b^{suff} replacing their invariance counterparts. This estimator preserves the expected gradient of
 113 the full-sum loss while reducing the per-step cost from $2|\mathcal{M}| + 1$ forward passes to three total (one
 114 clean, one for $\hat{\mathcal{L}}_{\text{inv}}$, one for $\hat{\mathcal{L}}_{\text{suff}}$), independent of the number of modalities.